

Navneet Kishan Srinivasan

Software Engineer | AI/ML | Backend & Full-Stack | Cloud-Native | Systems Thinking

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Summary

Backend and AI engineer with hands-on Python experience building monitoring, observability, and containerised infrastructure for production systems. Built internal monitoring frameworks and Docker-based microservices, maintained production Python pipelines, and developed LLM integrations with retrieval systems and vector databases. Experienced with logging, alerting, debugging, incident investigation, API security, data privacy, and operational reliability, with growing hands-on experience in AWS and Infrastructure as Code. Continually explores new tools through personal AI and API projects, applying them to improve workflows and deliver measurable impact.

Skills

Languages: Python, SQL, JavaScript, C#, Java

Platform & Infrastructure: Docker, Docker Compose, Linux, Git, CI/CD, AWS (fundamentals), Infrastructure as Code (fundamentals), Microservices, Cloud-Native Architecture

Observability & Reliability: Application Monitoring, Logging & Log Analysis, Alerting, Incident Management, Root Cause Analysis, Debugging & Troubleshooting, Dashboards, Telemetry, Production Reliability

AI/LLM: LLM APIs, LLM Integration, LangChain, RAG, FAISS, Vector Databases, Agentic Workflows, LLM Cost Management, Token Budgeting, PyTorch, TensorFlow, MCP Servers

Backend & Security: FastAPI, REST APIs, Node.js, MongoDB, pytest, API Authentication, API Access Control, Role-Based Access Control, Data Privacy | **Spoken:** English (C2), German (A2)

Work Experience

Applied AI/ML Engineer

02/2026 – Present

Siemens Healthineers, Erlangen, Germany · Working Student

- Develop and maintain production Python data pipelines for MRI in the Diagnostic Imaging Digital & Automation Platforms team, debugging and extending existing codebases.
- Reduced end-to-end pipeline runtime by 30% through processing optimisations, and contributed to scaling the pipeline for larger datasets.
- Investigate pipeline issues through debugging and log analysis to identify root causes and improve operational reliability.

Quantitative Developer

06/2024 – 03/2025

Citi, Budapest, Hungary · Internship

- Rebuilt an internal monitoring and observability framework with dashboards for a distributed production system, improving visibility into capacity, utilisation hot spots, and reliability.
- Developed and maintained backend services in Python, MongoDB, REST APIs, and Node.js within an existing production codebase.
- Wrote pytest-based test suites to catch regressions before release, supporting release validation across teams.
- Investigated incidents through debugging and root-cause analysis in an Agile team, collaborating with engineers to identify and implement fixes efficiently.

Programming Instructor – Python & OOP / Java

02/2025 – 06/2025

ELTE, Budapest, Hungary · Part-time

- Taught two undergraduate courses (weekly labs, 30+ students) covering SOLID, UML, and software design; supported 140+ students with debugging and troubleshooting.

Technical Lead, Google Developers Group on Campus

10/2024 – 06/2025

Google Developers Group, Budapest, Hungary · Part-time

- Built automated evaluation frameworks for coding challenges used in workshops and hackathons.

Projects

MedLedger – Full Stack Containerised Data-Processing Platform *Python, FastAPI, Docker, MongoDB, HAPI FHIR*

- Deployed a modular platform as Docker-based microservices with Docker Compose; debugged service-to-service communication, environment isolation, and reproducible deployments.
- Implemented anomaly-detection routines flagging abnormal workload behaviour, plus role-based access control, blockchain auditable and schema-driven validation for sensitive medical data – governance, security, and data-privacy controls in practice.

PaperLens – Cost-Capped Autonomous LLM Agent · *team project*

Python, Claude API, Agentic Workflows

- Built an agentic pipeline that extracts claims from papers, provisions repositories, proposes evaluation commands via LLM, and self-corrects on failure through an automated retry loop.
- Implemented an LLM client layer with configurable model and backend selection, token budgeting, and hard per-run spend caps – direct experience with LLM API cost management and usage control.
- Shipped a real-time dashboard surfacing run status, pass rates, and estimated spend, giving visibility into pipeline reliability and LLM inference cost.

Education

MSc in Data Science – Specialising in AI/ML

2025 – 2027

FAU Erlangen-Nürnberg, Germany

BSc in Computer Science

2022 – 2025

Eötvös Loránd University (ELTE), Budapest, Hungary

Thesis: MedLedger – Secure Health Record System (Grade 1.0); special mention, TDK Scientific Conference at Siemens Mobility.